Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · www.solaress.org

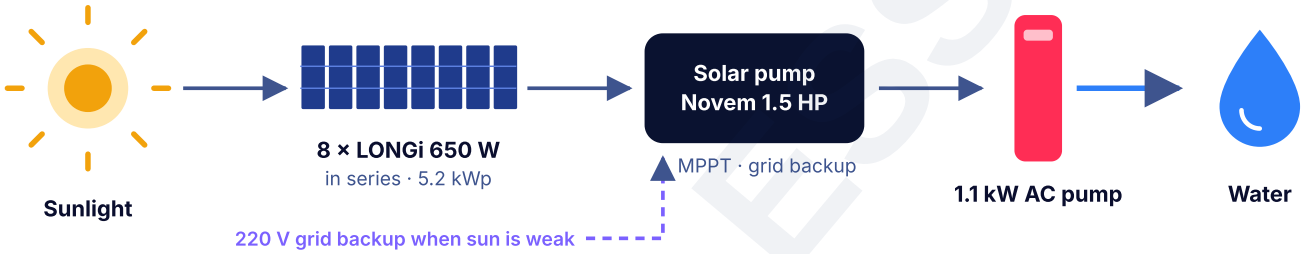
SOLAR WATER PUMPING · SYSTEM 1 · BUDGET BOM & PRICE

Hilltop tank system

Borehole 30–40 m, water pumped 30 m uphill over 200 m into a tank on the hill. A TORQUE 1.1 kW borehole pump, sized for this head, is driven by a Novem ND1-2-003-S solar pump inverter from 8 × LONGi 650 W panels, with 220 V grid power as backup when the sun is weak.

TN-20261019-PUMP-03-S1 | Rev07 | 20 October 2026 | Chiang Rai, Thailand | Prices in THB, confirmed at order

How it works



1 Hilltop tank

Borehole 30–40 m, 200 m uphill pipe to a tank 30 m above the well.

≈ 70 m	12–18 m³	200 m	2–3 m³/h
Total dynamic head	Water per sunny day*	Uphill pipe run	Expected flow*

SYSTEM 1 PRICE
120,850 THB excl. VAT
129,310 THB including 7% VAT

The pump · TORQUE TQ-SP-2DLBM20-4/S

1.1 kW 1.5 HP motor	**220 V** 1-phase AC	**130 m** Max head	**53–119 m** Working head range
1½" Outlet	**20** Impellers · 4" well	**8 × 650 W** LONGi panels · 5.2 kWp	**Novem ND1-2-003-S** 1.5 HP · 11 A pump inverter

* Flow and daily volume are estimates: the maker does not publish flow figures for this model. They will be confirmed with the pump curve at 70 m before the final quotation.

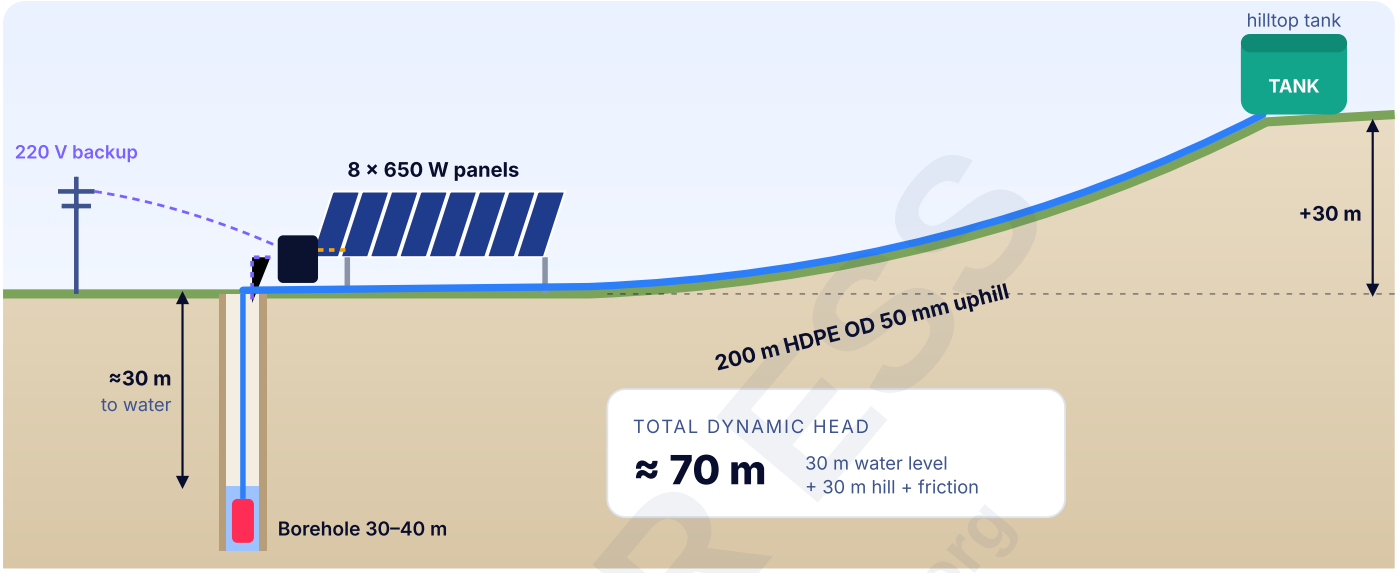
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SYSTEM 1 · THE SITE

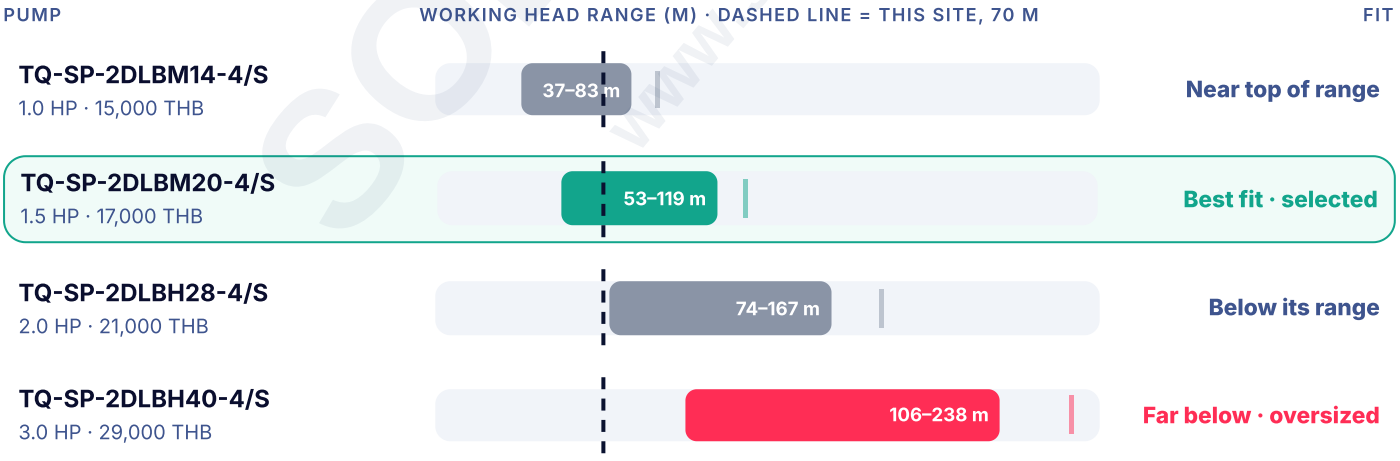
Borehole 30–40 m, pumped 30 m uphill over 200 m

Total dynamic head ≈ 70 m (water level ~30 m + 30 m hill + friction). Expected output about 2–3 m³/h into the hilltop tank, roughly 12–18 m³ per sunny day, to be confirmed with the pump curve.



HEAD	FLOW	PER SUNNY DAY	PUMP CURVE
≈ 70 m	2–3 m³/h	12–18 m³	To confirm
water level + hill + friction	estimate, at ≈ 65–70 m head	12,000–18,000 L	flow at 70 m before final quote

Why this pump: which TORQUE pump works best at 70 m?



Each bar is the head range the maker recommends for that pump; the thin line is its maximum. The 1.5 HP TQ-SP-2DLBM20-4/S is the only one with 70 m comfortably inside its range: enough reserve for the hill and the 200 m pipe, without the larger pumps running off the end of their curves and wasting power.

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SYSTEM 1 · BILL OF MATERIALS

Hilltop tank system · price

TOTAL EXCL. VAT	VAT 7%	TOTAL INCL. VAT	OPTIONAL TANK
120,850 THB	**8,460** THB	**129,310** THB	**≈ 9,500** 3,000 L PE + base, not in total

Where the money goes



Pump & inverter **27,500** · Solar array & frame **46,400** · Piping & valves **15,300** · Electrical & protection **10,850**
Safety accessories **800** · Installation **20,000**

Itemised list

#	ITEM	QTY	UNIT (THB)	AMOUNT (THB)
1	TORQUE TQ-SP-2DLBM20-4/S borehole pump, 1.1 kW (1.5 HP), 1-phase 220 V, 1½" outlet, 20 impellers, for 4" well	1 set	17,000	**17,000**
2	Novem ND1-2-003-S solar pump inverter, 1.5 HP 1-phase 220 V output, 11 A, MPPT 250–450 VDC, hybrid PV + 220 V grid input, dry-run protection (budget price)	1	10,500	**10,500**
3	LONGi 650 W solar panel	8	3,800	**30,400**
4	Galvanised ground-mount frame, 8 panels	1 set	16,000	**16,000**
5	HDPE pipe OD 50 mm PN10 (well drop + 200 m uphill line)	240 m	45	**10,800**
6	HDPE fittings, 1½"→OD50 adaptor, check valve, gate valve	1 lot	4,500	**4,500**
7	Submersible pump cable 4 × 2.5 mm²	50 m	55	**2,750**
8	PV cable 4 mm² red/black + MC4 connectors	1 lot	2,000	**2,000**
9	DC breaker + DC SPD + steel enclosure	1 set	3,500	**3,500**
10	AC breaker for grid backup supply	1	800	**800**
11	Earth rod and earth cable	1 set	1,200	**1,200**
12	Tank float switch	1	600	**600**
13	Stainless safety rope + pump clamps + well cap	1 set	800	**800**
14	Installation, pipe laying and commissioning	1 lot	20,000	**20,000**
	Total (excluding VAT)			**120,850**
	VAT 7%			8,460
	Total including VAT			**129,310**

Notes

Prices are budget figures excluding VAT unless stated. Pump and inverter prices are from current market listings; other items are market estimates. Final prices are confirmed at order.

The maker publishes maximum head (130 m) and working range (53–119 m) but no flow figures. At about 70 m the pump sits near the middle of its range, where a borehole pump is most efficient. The 2–3 m³/h estimate comes from the motor power and head; the pump curve will be obtained to confirm it before the final quotation.

The Novem ND1-2-003-S (11 A) is rated for 1.5 HP single-phase 220 V motors, matching the 1.1 kW pump (about 7–8 A). 8 × 650 W in series (≈ 300 V MPP, ≈ 395 V Voc) stays inside its 250–450 VDC range even on hot days, and the extra array power starts the pump earlier and runs it longer. Its price and how it switches to grid power are confirmed with Novem before order.

All electrical work uses copper cable, proper DC protection and earthing, installed by a certified electrician.